

FlowCIR: Semantic Transport via Flow Matching for Zero-Shot Composed Image Retrieval

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Abstract. Zero-shot composed image retrieval (ZS-CIR) aims to retrieve a target image by editing a reference image with a natural-language instruction, without relying on domain-specific annotated triplets. Most existing ZS-CIR methods perform composition in the text space, *e.g.*, by translating the reference image into pseudo-text tokens and applying token-level concatenation, which can be lossy and brittle for fine-grained semantics. In this work, we propose a new paradigm that casts ZS-CIR as *conditioned semantic transport* in the embedding space. Leveraging *conditional flow matching*, our model learns a lightweight transport field that maps the instruction representation toward a target-aligned query embedding conditioned on the reference image. The resulting framework is training-efficient, requiring roughly $10\times$ fewer training resources than prior textual-inversion-based approaches. We further identify negation and removal as a major failure mode of CLIP-based composition and introduce a multi-negative-aware strategy that repels the transported embedding from multiple undesired semantic anchors, improving robustness on negation-heavy queries. Extensive experiments on standard CIR benchmarks demonstrate consistent improvements over strong ZS-CIR baselines.

Keywords: Composed Image Retrieval · Flow Matching · Semantic Transport

1 Introduction

Composed image retrieval (CIR) [5, 10, 15, 33, 42] offers a practical visual search interface: rather than describing a target image from scratch, a user provides a reference image and a relative instruction, and the system retrieves images that match the intended edit (See Fig. 1(a)). Previous approaches are supervised and rely on human-annotated triplets whose large-scale collection is laborious and time-consuming. In addition, models trained on a particular dataset or domain often generalize poorly beyond their training distribution. These limitations motivate the zero-shot setting [4, 40], which studies CIR without domain-specific annotations and typically relies on unsupervised objectives or synthetic supervision to learn more transferable compositional representations.

Recent zero-shot CIR methods [4, 17, 40, 41] largely build on textual inversion over well-aligned vision-language models [26, 37], effectively reducing cross-modal composition to a text-only manipulation problem. Concretely, they learn

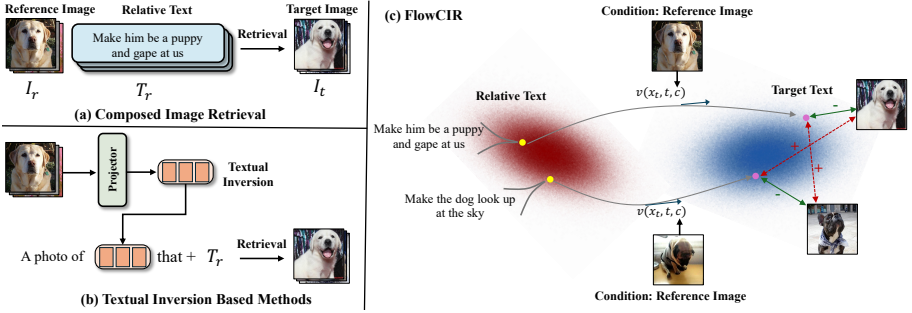


Fig. 1: (a) Illustration of Composed Image Retrieval. (b) Prior methods are composed by textual inversion and token-level fusion in text space. (c) FlowCIR composes via conditional flow matching to produce a target-oriented retrieval query.

a projector to *transfer* the reference image I_r into a small set of pseudo-text tokens and compose them with the relative instruction T_r in the text space to form a retrieval query, which is then matched against gallery image embeddings as shown in Fig 1(b). This paradigm rests on the key assumption that the semantics of an image can be faithfully captured by a few textual tokens and manipulated entirely within a textual space. However, this assumption is often impractical: compressing a rich reference image into only a handful of tokens inevitably loses fine-grained visual cues, so the composed representation may miss essential semantics needed for accurate retrieval. Moreover, learning an effective inversion projector often requires substantial computation, typically involving hours of large-batch contrastive training on multiple GPUs with web-scale image-text data [16, 17, 40]. These limitations motivate a more reliable and training-efficient cross-modal composition mechanism beyond tokenizing images and operating purely in text space.

In this work, we utilize *conditional flow matching* to reframe zero-shot CIR as a *semantic transport* problem in the embedding space. The relative instruction T_r conveys the desired semantic shift, while the reference image I_r provides the visual content to be preserved and serves as the conditioning context for the transformation. Rather than constructing the retrieval query via a single, static concatenation in the text space, we perform composition as a *continuous* semantic transformation via *conditional flow matching*, mitigating the semantic loss from textual inversion and enabling a conditioned, progressive composition process

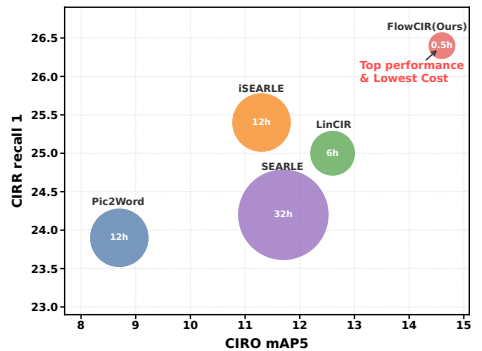


Fig. 2: Performance on CIRCO (mAP@5) and CIRR (Recall@1) together with training hours (on single GPU).

guided by I_r rather than a single concatenation. Formally, we learn a conditional transport that maps relative-instruction embeddings to target-oriented text embeddings, with the reference image embedding as the conditioning context. This formulation provides a principled mechanism for cross-modal composition and is also training-efficient: we only learn a lightweight conditional velocity field, enabling training within about one hour on a single GPU. As summarized in Fig. 2, our method achieves strong performance on both CIRR and CIRCO while incurring substantially lower training cost than prior baselines.

Meanwhile, as our transport is initialized from the instruction embedding, it may be influenced by a known weakness of CLIP-style VLMs: negation is often geometrically ambiguous in the text space [3, 38]. For example, expressions such as “a dog” and “remove/no dog” may lie undesirably close, providing an unreliable starting signal and biasing the transfer toward the negated concept. Therefore, inspired by [38], we propose an inference-only *multi-negative aware* steering strategy that explicitly pushes the composed instruction representation away from negative semantic directions, alleviating affirmative/negated collapse in CLIP space and improving retrieval on negation-heavy queries.

In summary, our contributions are three-fold. (i) We introduce a new paradigm for zero-shot composed image retrieval by conditioned semantic transport in embedding space. (ii) We propose a multi-negative aware extension that substantially improves robustness to negation and removal instructions, a prominent failure mode of CLIP-based composition. (iii) We achieve consistent improvements on standard CIR benchmarks, while remaining training-efficient.

2 Related Work

Zero-Shot Composed Image Retrieval Composed image retrieval (CIR) [42, 49] aims to retrieve target images given a reference image and a modification text describing the desired change. To avoid relying on heavily annotated real-world triplets, recent studies propose zero-shot CIR (ZS-CIR) [4, 40], where models perform composition and retrieval without task-specific manual annotations. A critical challenge in ZS-CIR is how to effectively compose the visual evidence from the reference image with the semantic shift implied by the text.

Existing ZS-CIR methods can be broadly grouped into three lines. (1) *Textual-inversion based methods* [4, 17, 40, 41] convert image information into learnable pseudo tokens, and then perform composition in the text embedding space for retrieval; representative works include Pic2Word [40] and SEARLE [4] and subsequent variants that design context-dependent mappings [41] or multiple pseudo words [11] to enhance controllability. (2) *Generative-based methods* introduce generative models (e.g., diffusion models) to synthesize target-oriented features [16] or leverage large-scale pretrained image generators (e.g., Stable Diffusion [39]) to produce pseudo target images as auxiliary visual evidence [43]. (3) *MLLM-based training-free methods* leverage large vision-language models to obtain richer textual signals without task-specific training: they first caption the reference image with a VLM (e.g., BLIP/Qwen-VL) and then use an

LLM/MLLM to parse, rewrite, or decompose the modification instruction and fuse it with the caption to form a composed query for retrieval [23, 28, 48], at the cost of extra inference-time computation.

In contrast to the above paradigms, we formulate composed image retrieval as *conditional semantic transport* and instantiate composition with conditional flow matching, which directly fuses the reference image evidence and relative instruction via a learned transport field rather than token inversion, image synthesis, or inference-time prompting.

Flow Matching (FM) Flow Matching (FM) has recently gained momentum as a powerful alternative for image generation. It learns a time-conditioned velocity field (a neural ODE) that transports samples along a predefined probability path from a simple prior to the data distribution by directly regressing the corresponding velocities [2, 29, 32]. Early studies [12, 14] instantiate this transport from noise to images, achieving high-fidelity synthesis on par with diffusion models [20] while retaining simpler training and sampling dynamics. More recent text-to-image FM variants [19, 30] further learn cross-modal transports from text embeddings to image embeddings, enabling direct language-to-vision generation through learned velocity fields. Beyond text-conditioned synthesis, FM has also been adapted to visual domain transfer problems such as monocular depth estimation [18] and semantic segmentation [6, 46], by formulating them as transport between distributions defined over different visual representations.

Generative Models for Perception Beyond synthesis, modern generative models have become increasingly useful for visual recognition and perception tasks [7, 8, 25, 31, 36, 44, 45]. One prominent direction leverages generators as *data engines*, producing realistic and diverse samples to mitigate limited supervision and improve performance in fine-grained recognition [13, 22], few-shot learning [44], long-tailed classification [24], and category discovery [31]. In parallel, a complementary line uses the generative process for recognition itself: text-to-image diffusion models can be repurposed as zero-shot or few-shot classifiers by evaluating class-conditioned denoising likelihoods (diffusion classifiers) [7, 8, 25], with efficiency improved via hierarchical prompting [35] and auxiliary image encoders [36]. More recent studies further explore parameterizing few-shot adaptation through diffusion time steps to capture subtle attributes [50], and selecting or optimizing the injected noise to stabilize and strengthen the recognition pipeline [45].

3 Preliminary

3.1 Flow Matching

Flow Matching [2, 29] aims to learn a continuous-time transport between two distributions by regressing a time-dependent velocity field. Under the rectified

flow instantiation [32], we define a linear path between paired samples ($\mathbf{x}_0 \sim X_0, \mathbf{x}_1 \sim X_1$): $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$ where $t \in [0, 1]$. The corresponding ground-truth velocity along this path is constant: $\mathbf{v}^*(\mathbf{x}_t, t) = \mathbf{x}_1 - \mathbf{x}_0$. Flow matching trains a neural field $\mathbf{v}_\theta(\mathbf{x}_t, t)$ to approximate this velocity. Concretely, we sample $(\mathbf{x}_0, \mathbf{x}_1)$ and draw $t \sim \mathcal{U}[0, 1]$ to construct $\mathbf{x}_t$, and minimize the flow-matching objective

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{\mathbf{x}_0, \mathbf{x}_1, t} \left[\left\| \mathbf{v}_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|_2^2 \right], \quad (1)$$

which is equivalent to matching the marginal velocity field in the original flow-matching formulation [29, 32]. Once trained, samples can be transported between the two distributions by solving the ODE $\frac{d\mathbf{x}_t}{dt} = \mathbf{v}_\theta(\mathbf{x}_t, t)$ from $t = 0$ to $t = 1$ (or in reverse).

For conditional flow matching, the data distribution $p(x)$ is a mixture (i.e., a marginal) over multiple class-conditional distributions, such as the categories in ImageNet. Its training is almost identical to standard flow matching, except that the velocity field is additionally conditioned on the target class label y . Accordingly, it learns a class-conditional vector field $\mathbf{v}_\theta(\mathbf{x}_t, t, y)$ by minimizing

$$\mathcal{L}_{\text{cond}}^{(\text{FM})}(\theta) = \mathbb{E} \left[\left\| \mathbf{v}_\theta(\mathbf{x}_t, t, y) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|_2^2 \right]. \quad (2)$$

In this work, we adopt the conditional flow matching formulation and use linear interpolation paths to transport relative instruction embeddings toward the target text embedding conditioned on the reference image embedding.

4 Method: FlowCIR

4.1 Problem Statement

Given a database of candidate images $\mathcal{D} = \{I_i\}_{i=1}^N$, zero-shot composed image retrieval (ZS-CIR) considers a query triplet $\langle I_r, T_r, I_t \rangle$, where I_r is a reference image, T_r is a relative text instruction describing how the target should differ from I_r , and I_t is the (unknown) target image that satisfies the instruction. The goal is to effectively compose cross-modal information from the reference image I_r and the relative text T_r to retrieve the correct target image I_t from $\mathcal{D}$. In the *zero-shot* setting, the model is trained without using CIR-specific annotated triplets $\langle I_r, T_r, I_t \rangle$ from the target benchmarks; instead, it relies on unsupervised training on large-scale image-text pairs and/or training with synthesized data to generalize to unseen queries at test time.

4.2 Conditional Flow Matching for Semantic Transport

We train in a zero-shot manner using synthesized compositions of the form $\langle I_r, T_r, T_t, I_t \rangle$, where I_r is a reference image, T_r is a relative instruction, and (T_t, I_t) describes the target side. Given such a tuple, our goal is to learn a composed query embedding that retrieves I_t from the gallery. We extract frozen, ℓ_2 -normalized vision-language embeddings with a pretrained image encoder $E_I(\cdot)$

and a text encoder $E_T(\cdot)$ to obtain $\mathbf{x}_{I_r} = \frac{E_I(I_r)}{\|E_I(I_r)\|} \in \mathbb{R}^d$, $\mathbf{x}_{T_r} = \frac{E_T(T_r)}{\|E_T(T_r)\|} \in \mathbb{R}^d$, $\mathbf{x}_{T_t} = \frac{E_T(T_t)}{\|E_T(T_t)\|} \in \mathbb{R}^d$, and $\mathbf{x}_{I_t} = \frac{E_I(I_t)}{\|E_I(I_t)\|} \in \mathbb{R}^d$. Our model then learns a conditional semantic transport mapping $(\mathbf{x}_{T_r}, \mathbf{x}_{I_r}) \mapsto \mathbf{x}_{T_t}$ within the embedding space.

We utilize conditional flow matching to model composition as semantic transport in the embedding space. Given a tuple and the extracted features, we sample a time scalar $t \sim \mathcal{U}(0, 1)$ and construct the interpolated state along a linear path

$$\mathbf{x}_t = (1 - t)\mathbf{x}_{T_r} + t\mathbf{x}_{T_t}. \quad (3)$$

Our flow matching network f_θ predicts a time-dependent velocity conditioned on the reference image,

$$\hat{\mathbf{v}}_\theta = f_\theta(\mathbf{x}_t, \mathbf{x}_{I_r}, t), \quad (4)$$

and is trained to match the ground-truth constant velocity $\mathbf{v}^* = \mathbf{x}_{T_t} - \mathbf{x}_{T_r}$ via the flow-matching regression objective

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_t \left[\|\hat{\mathbf{v}}_\theta(\mathbf{x}_t, \mathbf{x}_{I_r}, t) - (\mathbf{x}_{T_t} - \mathbf{x}_{T_r})\|_2^2 \right]. \quad (5)$$

Besides regressing the velocity, we adopt a contrastive retrieval objective where the paired $(\hat{\mathbf{x}}_{T_t}, \mathbf{x}_{I_t})$ forms the positive match, and we mine top- K hard negatives from the remaining target-image embeddings in the minibatch to sharpen discrimination against challenging distractors. We first form a predicted target text embedding $\hat{\mathbf{x}}_{T_t}$ by a one-step transport from time t to 1,

$$\hat{\mathbf{x}}_{T_t} = \mathbf{x}_t + (1 - t)\hat{\mathbf{v}}_\theta(\mathbf{x}_t, \mathbf{x}_{I_r}, t), \quad (6)$$

and compute a sampled InfoNCE loss over the positive and the selected hard negatives. Specifically, we define the hard-negative index set as the top- K most similar candidates to the query (excluding the positive):

$$\mathcal{H}_K(\hat{\mathbf{x}}_{T_t}) = \text{TopK} \left(\left\{ \left\langle \frac{\hat{\mathbf{x}}_{T_t}}{\|\hat{\mathbf{x}}_{T_t}\|}, \mathbf{x}_{I_t}^j \right\rangle \right\}_{j \neq i} \right), \quad (7)$$

where $\text{TopK}(\cdot)$ returns the indices of the K largest similarities in the minibatch. The resulting top- K InfoNCE objective is

$$\mathcal{L}_{\text{RET}} = -\log \frac{\exp \left(\left\langle \frac{\hat{\mathbf{x}}_{T_t}}{\|\hat{\mathbf{x}}_{T_t}\|}, \mathbf{x}_{I_t} \right\rangle / \tau \right)}{\exp \left(\left\langle \frac{\hat{\mathbf{x}}_{T_t}}{\|\hat{\mathbf{x}}_{T_t}\|}, \mathbf{x}_{I_t} \right\rangle / \tau \right) + \sum_{j \in \mathcal{H}_K(\hat{\mathbf{x}}_{T_t})} \exp \left(\left\langle \frac{\hat{\mathbf{x}}_{T_t}}{\|\hat{\mathbf{x}}_{T_t}\|}, \mathbf{x}_{I_t}^j \right\rangle / \tau \right)}, \quad (8)$$

where $\mathbf{x}_{I_t}^j$ denotes the target-image embedding of the j -th sample in the minibatch, τ is a temperature scalar, and $\mathcal{H}_K(\hat{\mathbf{x}}_{T_t})$ indexes the K hardest negatives. The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\text{FM}} + \lambda \mathcal{L}_{\text{RET}}. \quad (9)$$

During inference, we adopt the same one-step transport scheme for efficiency: sampling t and predicting the velocity $f_\theta(\mathbf{x}_t, \mathbf{x}_{I_r}, t)$ yields direct one-step transport to $\hat{\mathbf{x}}_{T_t}$, which is then used to retrieve images by nearest-neighbor search in the gallery embedding space.

4.3 Multi-Negative Steering Strategy

Negation-heavy instructions (e.g., “no dog”, “remove stripes”) often induce ambiguous geometry in contrastive text spaces, where affirmative and negated descriptions can collapse to nearby directions. To address this issue without additional training, we propose an inference-only multi-negative steering strategy. During inference, given a relative instruction T_r , we decompose it into an affirmative intent T_r^a and a set of negated concepts $\{T_{r,k}^n\}_{k=1}^K$ using a lightweight parser (e.g., a small LLM or simple rule-based heuristics), and encode them as normalized embeddings $\mathbf{x}_{T_r^a} = E_T(T_r^a)/\|E_T(T_r^a)\|$ and $\mathbf{x}_{T_{r,k}^n} = E_T(T_{r,k}^n)/\|E_T(T_{r,k}^n)\|$. Our goal is to construct a *steered instruction* embedding $\hat{\mathbf{x}}_{T_r}$ that remains aligned with $\mathbf{x}_{T_r^a}$ while being repelled from all $\{\mathbf{x}_{T_{r,k}^n}\}_{k=1}^K$.

Following SpaceVLM [38], we model each concept as a neighborhood (spherical cap) in the VLM space: for a normalized point $\mathbf{x}$ and a cosine threshold $t \in [-1, 1]$,

$$\mathcal{N}(\mathbf{x}) = \{\mathbf{z} \in \mathbb{R}^d \mid \mathbf{x}^\top \mathbf{z} \geq t\}. \quad (10)$$

Accordingly, a feasible negation-aware representation should lie close to the affirmative intent but outside the neighborhoods induced by negatives, yielding

$$\mathcal{N}(T_r) = \mathcal{N}(\mathbf{x}_{T_r^a}) \cap \bigcap_{k=1}^K \mathcal{N}^c(\mathbf{x}_{T_{r,k}^n}), \quad (11)$$

where $\mathcal{N}^c(\cdot)$ denotes the complement. To obtain a single embedding compatible with our transport model, we extend this construction by computing, for each negative $\mathbf{x}_{T_{r,k}^n}$, a closed-form “center” direction:

$$\tilde{\mathbf{d}}_k = \mathbf{x}_{T_r^a} \frac{\sin(\alpha + \frac{\theta_k}{2})}{\sin \theta_k} - \mathbf{x}_{T_{r,k}^n} \frac{\sin(\alpha - \frac{\theta_k}{2})}{\sin \theta_k}, \quad \mathbf{d}_k = \frac{\tilde{\mathbf{d}}_k}{\|\tilde{\mathbf{d}}_k\|}, \quad (12)$$

where $\alpha = \arccos(\delta)$ is set by a cosine-margin hyperparameter $\delta \in [-1, 1]$ controlling the neighborhood size, and $\theta_k = \arccos(\mathbf{x}_{T_r^a}^\top \mathbf{x}_{T_{r,k}^n})$. When θ_k is very small, we use $\tilde{\mathbf{d}}_k = \mathbf{x}_{T_r^a} - \mathbf{x}_{T_{r,k}^n}$ for numerical stability. We then aggregate and normalize these directions to form the steered instruction embedding:

$$\hat{\mathbf{x}}_{T_r} = \frac{1}{K} \sum_{k=1}^K \mathbf{d}_k, \quad \hat{\mathbf{x}}_{T_r} \leftarrow \frac{\hat{\mathbf{x}}_{T_r}}{\|\hat{\mathbf{x}}_{T_r}\|}, \quad (13)$$

and replace the original instruction embedding with $\hat{\mathbf{x}}_{T_r}$ in our composition/transport pipeline, which improves robustness by separating affirmative and negated semantics in CLIP space.

Table 1: Comparison on CIRR (R@1/5/10) and CIRCO (mAP@5/10/25/50) under CLIP ViT-B/32 and ViT-L/14 backbones, including training cost (on single GPU).

Backbone	Method	Training(h)	CIRR			CIRCO			
			R@1	R@5	R@10	mAP@5	mAP@10	mAP@25	mAP@50
CLIP-ViT-B/32	Image-only	–	6.7	23.0	59.2	1.5	1.9	2.3	2.6
	Text-only	–	21.8	45.2	57.4	2.5	2.6	2.9	3.1
	PALAVRA [9]	–	16.6	43.5	58.5	4.6	5.3	6.3	6.8
	SEARLE [4]	32	24.3	53.3	66.1	8.9	9.4	10.6	11.2
	iSEARLE [1]	12	25.2	<u>55.7</u>	<u>68.1</u>	<u>10.6</u>	<u>11.2</u>	<u>12.5</u>	<u>13.3</u>
	SEARLE + CIG [43]	NA	<u>25.3</u>	54.8	<u>68.1</u>	10.2	10.6	11.8	12.5
	Ours	0.5	25.5	56.5	69.8	13.1	13.4	14.6	15.3
CLIP-ViT-L/14	Image-only	–	7.3	23.0	33.3	2.5	3.1	3.9	4.4
	Text-only	–	20.9	44.0	55.4	3.3	3.7	4.1	4.4
	Pic2Word [40]	128	23.9	51.7	65.3	8.7	9.5	10.6	11.3
	SEARLE [4]	32	24.2	52.4	66.3	11.7	12.7	14.3	15.1
	Context-I2W [41]	192	<u>25.6</u>	<u>55.1</u>	68.5	–	–	–	–
	LinCIR [17]	4	25.0	53.3	66.7	12.6	<u>13.6</u>	15.0	15.9
	LinCIR + CIG [43]	NA	<u>25.6</u>	54.8	67.6	<u>13.0</u>	<u>13.6</u>	15.1	16.0
	Compo-Dif [16]	29568	18.2	53.1	70.8	12.6	13.4	<u>15.8</u>	<u>16.4</u>
	iSEARLE [1]	12	25.4	54.1	67.5	11.3	12.7	14.5	15.3
	Ours	0.5	26.2	56.1	<u>68.6</u>	14.9	15.7	17.3	18.2

5 Experiments

5.1 Setups and Implementations

Datasets. Following prior zero-shot CIR works [1, 4, 40], we evaluate on three standard benchmarks: Fashion-IQ [47], CIRR [34], and CIRCO [4]. Fashion-IQ is an e-commerce fashion benchmark, where queries pair a product image with a brief modification text to retrieve the edited item. CIRR consists of real-world image pairs with relative captions describing the desired transformation from a reference to a target. CIRCO further scales composed retrieval to diverse everyday objects and scenes with more challenging distractors.

Implementation Details. Following prior zero-shot CIR works, we use CLIP [37] as the frozen vision–language backbone and report results with two variants: ViT-B/32 and ViT-L/14. For the flow matching model, we adopt the lightweight network design from MAR [27], implemented as a deep residual MLP with timestep conditioning. We train our model on the HQ-Edit-200k [21] synthetic image-editing dataset, which contains reference images paired with editing instructions, the corresponding edited results, and associated textual descriptions, matching the supervision signals required by our transport-based formulation. Unless otherwise specified, all experiments (training and inference) are conducted on a single NVIDIA RTX 3090 GPU with 24GB memory. Further details on datasets, optimization, and hyperparameters are deferred to the supplementary material.

Table 2: Comparison on Fashion-IQ under CLIP ViT-B/32 and ViT-L/14 backbones, reported in Recall@10 and Recall@50 for Dress/Shirt/Toptee and their average.

Backbone	Methods	Fashion-IQ							
		Dress		Shirt		Toptee		Average	
CLIP-ViT-B/32	Image-only	3.9	10.8	7.5	14.0	6.2	13.4	5.9	12.7
	Text-only	13.6	31.8	20.3	35.3	20.2	40.5	18.0	35.9
	PALAVRA [9]	17.3	35.9	<u>21.5</u>	37.1	20.6	38.8	19.8	37.3
	SEARLE [4]	<u>18.2</u>	<u>38.6</u>	24.8	41.1	<u>25.6</u>	<u>46.2</u>	<u>22.9</u>	<u>42.0</u>
	Ours	24.8	41.8	19.1	<u>40.1</u>	26.4	47.8	23.4	43.2
CLIP-ViT-L/14	Text-only	18.3	30.1	13.6	30.0	17.4	33.9	16.4	31.3
	Image-only	10.7	19.9	4.5	12.2	8.4	16.5	7.8	16.2
	Pic2Word [40]	20.0	40.2	26.2	43.6	27.9	47.4	24.7	43.7
	SEARLE [4]	20.5	43.1	26.9	45.6	29.3	50.0	25.6	46.2
	Context-L2W [41]	<u>23.1</u>	<u>45.3</u>	29.7	48.6	<u>30.6</u>	<u>52.9</u>	27.8	<u>48.9</u>
	LinCIR [17]	20.9	42.4	<u>29.1</u>	<u>46.8</u>	28.8	50.2	<u>26.3</u>	46.5
	Ours	31.6	48.5	24.4	44.3	33.2	53.9	29.7	48.9

5.2 Main Results

Tab. 1 and Tab. 2 summarize the zero-shot CIR performance on CIRR, CIRCO and Fashion-IQ under CLIP ViT-B/32 and ViT-L/14 backbones, together with single-GPU training cost. Overall, our method consistently achieves the best accuracy across benchmarks while being substantially more training-efficient than prior approaches. **CIRR&CIRCO.** On CLIP ViT-B/32, our method improves CIRR retrieval to 25.5/56.5/69.8 in R@1/5/10 and boosts CIRCO to 9.99/10.40/11.32/11.92 in mAP@5/10/25/50, surpassing strong token-composition baselines such as SEARLE and recent variants (Table 1). The gains are even more pronounced with the stronger CLIP ViT-L/14 backbone: our method reaches 26.4 R@1 on CIRR and 14.56 mAP@5 on CIRCO, while maintaining strong performance at higher cutoffs (mAP@50 17.76). Notably, we achieve these improvements with only **0.5** hours of training on a single GPU, compared with hours to days for prior methods (e.g., 32h for SEARLE and 128h+ for Pic2Word), highlighting a markedly better performance–cost trade-off. **Fashion-IQ.** Table 2 shows that our method also generalizes well to the e-commerce domain of Fashion-IQ. With CLIP ViT-B/32, we obtain the highest average retrieval accuracy (R@10/R@50: 23.8/42.9), outperforming the best competing baseline. With CLIP ViT-L/14, we further set a new state of the art, achieving an average of 29.7 R@10 and 48.9 R@50, with consistent improvements across Dress, Shirt, and Toptee categories. Together, these results demonstrate that our flow-based transport not only improves composed retrieval accuracy across diverse datasets, but also reduces training cost by over an order of magnitude, making it practical for efficient zero-shot deployment.

5.3 Diagnostic Analysis

Component Analysis. We conduct ablations to quantify the contributions of the two key components in FLOWCIR: Conditional Flow Matching (CFM)

Table 3: Ablation studies of different components in *FlowCIR* on CIRR and CIRCO. We evaluate the contributions of Conditional Flow Matching (CFM) and Multi-Negative Steering (Neg-Steering).

	CFM	Neg-Steering	CIRR			CIRCO			
			R@1	R@5	R@10	mAP@5	mAP@10	mAP@25	mAP@50
Text-Only	✗	✗	20.9	44.0	55.4	3.3	3.7	4.1	4.4
(1)	✗	✓	21.1	45.7	58.0	2.5	2.8	3.2	3.4
(2)	✓	✗	26.1	53.8	67.0	13.4	14.1	15.6	16.4
FlowCIR	✓	✓	26.2	56.1	68.6	14.9	15.7	17.3	18.2

and inference-time Multi-Negative Steering (Neg-Steering). As shown in Tab. 3, we report results on CIRR and CIRCO by progressively adding these components on top of a text-only baseline. **First**, the text-only baseline already provides a reasonable zero-shot retrieval model, achieving 20.9/44.0/55.4 on CIRR R@1/5/10 and 3.30/3.65/4.08/4.38 on CIRCO mAP@5/10/25/50. However, its performance remains limited, indicating that directly using textual composition is insufficient for accurate cross-modal retrieval. **Second**, adding Neg-Steering alone (Row (1)) brings clear improvements over the text-only baseline on CIRR, showing that explicitly steering the instruction away from negated semantics already improves the compositional query. Nevertheless, the gains on CIRCO remain modest, suggesting that inference-time negation handling alone cannot fully resolve the composition problem without a learned transport mechanism. **Third**, introducing CFM (Row (2)) leads to a substantial performance jump across all metrics, improving CIRR R@1 from 20.9 to 26.1 and CIRCO mAP@5 from 3.30 to 13.41. Adding Neg-Steering on top of CFM further improves all metrics, yielding the full FLOWCIR model with the best overall results: 26.4/55.4/68.1 on CIRR and 14.56/15.32/16.90/17.76 on CIRCO. These results show that CFM is the primary source of performance gain, while Neg-Steering is complementary and further improves robustness, especially when combined with the learned transport model.

Flows vs. Regressor. We further compare our flow-matching formulation with a direct regressor that predicts the target embedding in a single shot. As shown in the bottom part of Table 4, replacing flow matching with direct regression leads to a clear performance drop across all metrics, especially on CIRCO, where mAP@5 decreases from 14.8 to 12.3 on the test set. In contrast, the flow-matching model consistently achieves the best results on both validation and test splits. This comparison shows that learning a time-conditioned transport field is more effective than directly regressing the final embedding, likely because flow matching provides a more structured optimization target and better captures the semantic transition from instruction to target under the reference condition.

Hyperparameter analysis. Table 4 studies two important hyperparameters in FLOWCIR: the weight λ of the retrieval objective and the hard-negative ratio K , where K is defined as the proportion of mined negatives relative to the minibatch size. Overall, the method is reasonably stable across a broad range of settings,

Table 4: Hyperparameter analysis of *FlowCIR* on CIRR and CIRCO. We study the influence of λ , and K .

Param.	CIRR				CIRCO			
	Val		Test		Val		Test	
	R@1	R@10	R@1	R@10	mAP@5	mAP@25	mAP@5	mAP@25
$\lambda = 0.1$	25.4	69.3	25.9	67.2	10.2	13.1	10.6	12.5
$\lambda = 0.3$	<u>26.2</u>	69.7	26.4	<u>68.1</u>	13.9	15.7	14.2	16.8
$\lambda = 0.5$	26.6	<u>69.6</u>	<u>26.2</u>	68.6	<u>14.8</u>	<u>16.9</u>	<u>14.9</u>	<u>17.3</u>
$\lambda = 0.7$	26.1	69.5	<u>26.2</u>	<u>68.5</u>	14.8	17.0	15.2	17.6
$\lambda = 0.9$	26.0	69.5	25.8	68.4	14.5	16.8	14.8	17.2
$K = \frac{1}{16}B$	<u>26.1</u>	<u>69.2</u>	25.8	55.8	13.9	<u>15.7</u>	<u>14.2</u>	<u>16.7</u>
$K = \frac{1}{8}B$	26.6	69.6	26.2	68.6	14.8	16.9	14.9	17.3
$K = \frac{1}{4}B$	25.8	58.8	25.5	<u>67.6</u>	13.5	15.2	13.7	16.0
<i>Regressor</i>	<u>23.7</u>	<u>67.6</u>	<u>23.8</u>	<u>66.1</u>	<u>11.9</u>	<u>14.1</u>	<u>12.3</u>	<u>14.3</u>
<i>FM</i>	26.6	69.6	26.2	68.6	14.8	16.9	14.9	17.3

but moderate values consistently work best. For λ , using a very small value (*e.g.*, $\lambda = 0.1$) leads to weaker CIRCO performance, indicating insufficient retrieval supervision, while overly large values do not bring further gains. The best overall trade-off is achieved around $\lambda = 0.5$, which yields the strongest or near-strongest results on both CIRR and CIRCO. For hard-negative mining, selecting too few negatives (*e.g.*, $K = \frac{1}{16}B$) or too many negatives (*e.g.*, $K = \frac{1}{4}B$) is suboptimal, whereas a moderate choice $K = \frac{1}{8}B$ gives the best overall performance. This suggests that an appropriate number of hard negatives is important: too few fail to provide sufficient discriminative pressure, while too many may introduce noisy or overly aggressive distractors.

Negation Analysis.

5.4 Qualitative Analysis

Effect of reference-conditioned transport. Fig. 3 illustrates that the learned transport is strongly conditioned on the reference image, even under the same relative instruction. In the first two examples, the instruction “make the dog look up at the sky” leads to different retrieval behaviors depending on the reference: when the reference depicts a bulldog-like dog, the retrieved results largely preserve the breed/appearance of that dog while changing the pose; when the reference is a fluffy white dog, the retrieved images instead remain visually consistent with that appearance and mainly modify the head orientation. A similar phenomenon appears in the third and fourth examples with the instruction “the outfit of the lady has changed from blue to red,” where the retrieved results preserve the person’s pose, body shape, and scene context while altering the clothing attribute. These cases suggest that our conditional flow does not merely follow the text direction; instead, it learns a reference-dependent velocity field that transports the query toward target semantics while retaining identity and context cues from the reference image.

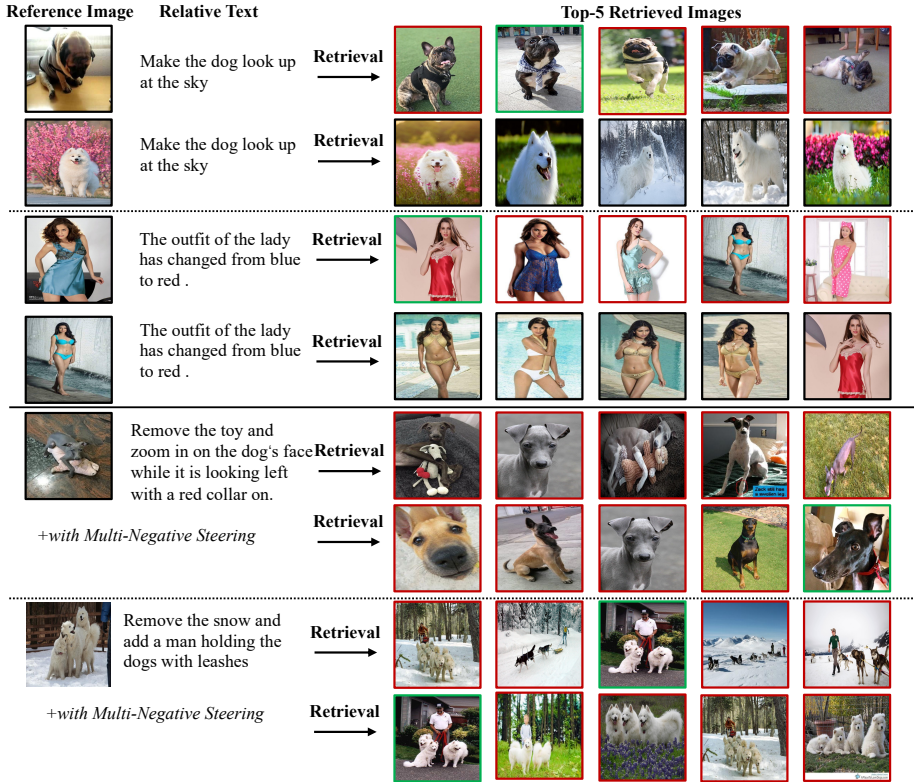


Fig. 3: Qualitative retrieval results illustrating the effect of reference-conditioned transport and Multi-Negative Steering. Predicted targets are marked with for correct retrievals and for incorrect ones.

Effect of multi-negative steering. The bottom examples in Fig. 3 highlight the benefit of Multi-Negative Steering on negation-heavy instructions. Without steering, retrieval often remains biased toward the negated concepts, reflecting the well-known difficulty of CLIP-style embeddings in separating affirmative and negated semantics. For instance, for “remove the toy and zoom in on the dog’s face while it is looking left with a red collar on,” the unsteered results still frequently contain distracting toys or irrelevant objects, whereas applying Multi-Negative Steering produces more dog-face-centered results and better respects the intended removal. Likewise, for “remove the snow and add a man holding the dogs with leashes,” the unsteered retrieval remains dominated by snowy scenes, while the steered version retrieves images with a human, leashes, and non-snowy environments more consistently. These examples show that Multi-Negative Steering effectively pushes the composed representation away from undesired semantic directions, thereby alleviating negation collapse and improving retrieval faithfulness.

6 Conclusion

In this paper, we presented FLOWCIR, a new paradigm for zero-shot composed image retrieval that formulates composition as conditioned semantic transport via flow matching in the embedding space. Instead of reducing cross-modal composition to static token operations in text space, our method learns a lightweight conditional velocity field that transports the instruction representation toward a target-aligned query embedding under the guidance of the reference image. To further address a key failure mode of CLIP-based composition, we introduced an inference-only Multi-Negative Steering strategy that pushes the composed representation away from undesired semantic directions and improves robustness on negation-heavy queries. Extensive experiments on Fashion-IQ, CIRR, and CIRCO demonstrate that FLOWCIR consistently outperforms prior zero-shot CIR methods while requiring substantially lower training cost. We hope this work can motivate future research on flow-based compositional retrieval and more principled modeling of negation and semantic control in vision-language retrieval.

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